Design and Development of an AI-Enabled Smart Electric Vehicle Charging and Energy Management Platform

Project Specification and Preliminary Report

Sub-project 1: EV Charging Data Analysis and Demand Forecasting

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I affirm that this submission is all my own work in accordance with the University of Glasgow and UESTC Regulations, and the James Watt School of Engineering requirements. *Signed:* _____ *Date:* _____

1 Project Description and Background

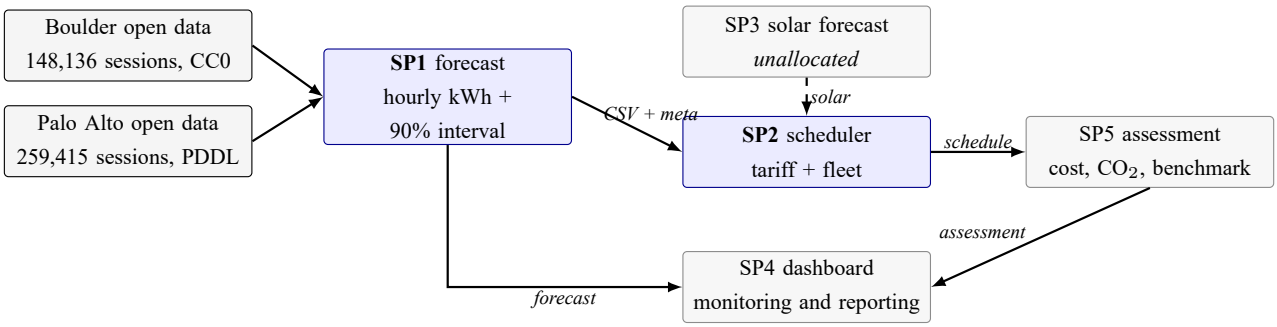


Figure 1: Platform architecture. Blue boxes are the components this report builds; grey boxes are the other sub-projects, shown only to fix the interfaces.

Electric vehicle (EV) adoption is adding load to distribution networks faster than network capacity is being reinforced, and the timing of that load is set by drivers rather than by the network. Uncoordinated charging concentrates demand in the hours when vehicles are parked and their owners are available to plug in, which raises peak demand, raises the price paid for energy, and can force curtailment of renewable generation that would otherwise have been absorbed. Coordinated charging inverts the problem: the same energy can be delivered at a different time, so peak demand falls without any reduction in the service provided. The platform in this project implements that inversion as a software pipeline: forecast the demand, schedule it against a tariff and a generation profile, display the result, and quantify what it saved.

The platform is split into five sub-projects with fixed interfaces. Sub-project 1 (SP1, this report) converts charging session records into an hourly demand series and forecasts the next 24 hours with a prediction interval. SP2 schedules that demand against a time-of-use tariff and a fleet of vehicles. SP3 supplies a solar generation forecast which SP2 may consume; it is not yet allocated to a student, and the interface is defined so that its absence is treated as zero solar rather than as a blocker. SP4 is the monitoring dashboard that consumes every other sub-project's output file. SP5 assesses the economic and environmental outcome. Figure 1 shows the data flow and the contract file carried on each edge.

The forecasting layer is the one that determines whether the rest of the platform can be evaluated at all: a schedule is only as good as the demand it is built on, and a dispatch constraint has to hold against the upper end of the forecast interval rather than against a point forecast. The importance is measurable rather than rhetorical. On the primary site used in this report, if every session charges as soon as it connects, the hourly peak reaches 362.5 kWh against a forecast

peak of 94.5 kWh, an amplification of 3.84, while total energy delivered moves by 0.05 %. The saving available to the platform therefore lies in moving load in time, not in reducing it, and the size of that saving is set by how well demand can be predicted and how tightly its uncertainty can be bounded.

2 Literature Review

2.1 Forecasting EV charging demand

Published work on station-level load forecasting divides into model-family comparisons and one systematised review, and the model family that wins depends on the data rather than on the architecture. Zhu et al. [1] compare four deep architectures on hourly station load and report a gated recurrent unit network as the best performer, framing the problem as a large fleet power draw driven by stochastic driver behaviour. Vishnu et al. [2] compare an auto-regressive model, support vector regression and an LSTM on Adaptive Charging Network data, report the LSTM as best with MAE 4 kW and RMSE 5.9 kW, and provide the most concrete published error figure in this strand. Zhang et al. [3] combine a multi-channel convolutional network with a temporal convolutional network and add meteorological and calendar covariates, improving on an ANN by 14.09 %, on an LSTM by 25.13 % and on a TCN by 4.48 %. Ge et al. [4] instead use a random forest with harmony-search tuning and construct the spatial-temporal load of vehicle clusters bottom-up. Alaraj et al. [5] survey the field and state the ordering that this project adopts: unmanaged adoption risks network instability, smart charging mitigates it, and smart charging presupposes an accurate forecast.

Table 1: Representative forecasting studies and what they establish for this project.

Study	Data and level	Model	What it settles here
Zhu et al. [1]	Station, hourly	GRU	Recurrent models are competitive at station level
Vishnu et al. [2]	ACN, station	LSTM	MAE 4 kW; a reference error figure from the same data family
Zhang et al. [3]	Urban station, 15 min	CNN + TCN	Calendar and weather covariates improve accuracy
Ge et al. [4]	Cluster	Random forest	Spatial aggregation needs a bottom-up construction
Alaraj et al. [5]	Review	—	Forecast first, schedule second

2.2 From session records to hourly load profiles

The project's central data-engineering assumption is that a table of sessions, each carrying a connection time, a disconnection time and a metered energy quantity, can be turned into an hourly power profile. Yi and Scoffield [6] estimate the distribution of charging duration conditional on parking duration and generate synthetic behaviour that preserves the parking-to-travel dependency. Uimonen and Lehtonen [7] derive office-building profiles from measured charger occupancy, the closest published analogue to the workplace site used here. Munkhammar et al. [8] reduce a Monte Carlo state-of-charge simulation to hour-by-hour Bernoulli distributions of plug-in status and show that the aggregate over many vehicles is approximately normal, which is the formal basis for turning per-vehicle plug-in indicators into a fleet profile. Lahariya et al. [9] model arrivals as a Poisson-like process with connection time dependent on arrival time, and von Hoffen [10] makes the complementary point that transaction records acquire analytical value only after enrichment with context. The consequence for this project is that charging duration is modelled as a function of occupancy rather than as a constant rate, which is why a session's metered energy is spread across its charging window in proportion to overlap instead of being divided by a rated power.

2.3 Tariffs, scheduling and peak reduction

Chen et al. [11] formulate joint optimal power flow and EV charging for a fleet with heterogeneous charging-rate limits and characterise the optimal offline schedule as a valley-filling profile, giving both a centralised and a decentralised online algorithm that tracks it; this is the benchmark against which a scheduling heuristic can be judged approximately optimal. Ramadan et al. [12] solve the related peak-shaving problem with particle swarm optimisation and allow vehicle-to-grid discharge during peaks. On the tariff side, Davis and Bradley [13] test whether time-of-use rates actually shift charging and find that for plug-in hybrids the extra fuelling cost of charging off-peak can cancel the incentive, so tariff response is not free and a cost-saving result that assumes it is free overstates the benefit. Szinai et al. [14] link mobility and dispatch models for California and report that 0.95–5 million managed vehicles avoid USD 120–690 million in annual operating costs, up to 10 %, and cut renewable curtailment by up to 40 %, while noting that overnight time-of-use charging increases curtailment at similar cost savings. Dixon et al. [15] schedule against grid carbon intensity and wind availability, reducing average charging emissions from 35–56 to 28–40 gCO_2/km. Cost and curtailment are therefore separately measurable objectives that can move in opposite directions, and a schedule has to state which one it optimises.

2.4 Open data and measured grid impact

Amara-Ouali et al. [16] survey open data for reproducible EV load research, searching more than 860 repositories and returning roughly 60 datasets, and motivate the survey by the field's shortage of charging-station data of realistic consistency. That finding framed the dataset choice made here: official municipal releases were selected on provenance, a public-domain or open licence, metered energy per session, and downloadable access without registration. Li et al. [17] publish UrbanEV, covering charging-space availability and electricity consumption in Shenzhen at hourly resolution for more than 20,000 stations, and Lee et al. [18] present ACN-Sim, which also serves as the DOI-bearing citation for the Adaptive Charging Network dataset. On the impact side, Powell et al. [19] evaluate charging demand against a 2035 economic dispatch for the Western Interconnection and find peak net demand rising by up to 25 % under forecast adoption and 50 % in a full-electrification stress test, with the counter-intuitive result that locally optimised charging combined with a high share of home charging can strain the grid, while uncontrolled *daytime* charging reduces storage, ramping and emissions. Verma et al. [20] review distribution-network impacts across penetration levels, supporting the treatment of impact as a function of fleet size rather than as a fixed property of EVs.

2.5 Significance and the gap addressed

Three conclusions follow from the review and shape the specification. First, one-hour resolution with a rolling-origin out-of-sample protocol and explicit naive baselines is the minimum design under which a reported accuracy means anything; the reviewed papers mostly compare model families against each other, and a learned model that does not beat persistence has demonstrated nothing. Second, published absolute errors come from other datasets and larger sites, so they are context rather than targets. Third, session-level open data with metered energy is scarce, and the Chinese official route was closed after verification: the national charging platform publishes monthly aggregates for members, not per-session records, so the forecasting work rests on two municipal releases.

What is missing in that literature, and what this project therefore specifies, is a forecasting layer built on open municipal session data that (i) reports accuracy against naive baselines rather than only against other learned models, (ii) carries a calibrated prediction interval that is evaluated by its coverage rather than assumed, and (iii) demonstrates that the result transfers to a second city without retuning. The next section turns the gap into objectives.

3 SMART Objectives

Four objectives are specified. Each states a measurable target and the date by which it is met, and each maps to a task in the work plan of Section 6.

O1. **A reproducible demand series from open session data.** By 30 October 2026, convert the two official municipal session datasets into gap-free hourly demand series in UTC, with a contract-validated CSV plus metadata sidecar and at least 50 automated tests covering ingestion, aggregation and time-zone handling. *Achieved: 77 tests pass; both series are produced by one command from a checksummed download.*

O2. **Day-ahead forecasts that beat persistence on both sites.** By 31 January 2027, produce 1–24 h forecasts that reduce WAPE against the naive same-hour-yesterday baseline by at least 15 % on the primary site and by at least 10 % on the validation site, with measured coverage of the nominal 90 % interval within ±3 percentage points. *Partially achieved: 19 % and 18 % reductions, coverage 91.0 % and 90.5 %.*

O3. **A quantified peak-shifting target for the scheduler.** By 30 April 2027, quantify, for the primary site, the peak reduction and cost difference between uncontrolled charging and a schedule built on the forecast, and deliver both series to SP5 for the assessment. *In progress: the uncontrolled baseline is built; the scheduled series depends on SP2.*

O4. **An interface that other sub-projects consume unchanged.** By 30 April 2027, deliver the forecast as a contract file that SP2, SP4 and SP5 read without transformation, with the interval columns populated and the method recorded in the sidecar. *Partially achieved: the contract file and sidecar exist and are versioned.*

The key contributions are the use of official municipal session data as a forecasting basis where the field mostly relies on proprietary or single-site records, a direct multi-horizon formulation in which the forecast step is an input feature rather than a recursion, and an interval that is evaluated by its coverage instead of being asserted.

4 Technical Background

4.1 Hypothesis and method

The hypothesis is that aggregate charging demand at a municipal or workplace site is governed by calendar structure and by the recent level of demand, so that a gradient-boosted model on target-hour calendar values plus forecast-origin lags, with the forecast horizon supplied as a feature, predicts one to twenty-four hours ahead more accurately than persistence at every horizon. Two consequences follow. Because the horizon is a feature, one model serves all horizons and needs no recursion, which removes the error compounding that a one-step model accumulates when iterated. Because every feature is either a calendar value of the *target* interval or a lag anchored at the *origin*, no information from the future can enter the model; a regression test recomputes each lag from the raw series and fails if any feature can see its own target.

4.2 Data and derivation

Two official municipal releases are used. The City of Boulder, Colorado, publishes 148,136 charging transactions at 50 city-owned Level-2 stations between January 2018 and December 2023, with metered energy per session, under CC0; it is the primary site and supplies 1.253 GWh. The City of Palo Alto, California, publishes 259,415 sessions at 47 stations between July 2011 and January 2021 under the PDDL, supplying 2.216 GWh; it is used only for validation. Both are session-level records with connection time, disconnection time and metered energy, which is the minimum schema the derivation needs.

Each session's metered energy is spread over the hourly intervals it actually overlaps, in proportion to the overlap, and the timeline is closed into a gap-free hourly grid. This is the occupancy-conditional treatment argued for by [6] and [7], with one limitation inherited from the source: the records carry no within-session power taper, so a constant rate is assumed inside a session. The assumption is recorded in the output metadata rather than left implicit.

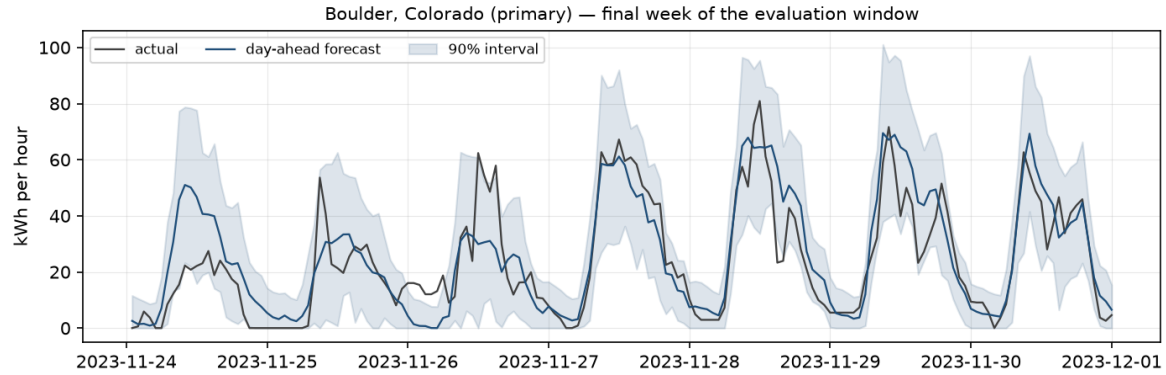


Figure 2: Metered demand, day-ahead forecast and the 90 % prediction interval over the final week of the evaluation window on the primary site. The band widens where demand is volatile because it is built from the residual quantiles of the backtest at each horizon.

4.3 Evaluation and interval construction

Seven models are scored: three naive baselines (same hour yesterday, same hour last week, and an hour-of-week profile) and four learned models (ridge regression, random forest, gradient boosting and XGBoost). Evaluation is a rolling-origin backtest in which every model is retrained at every origin using only targets strictly before that origin, and each origin forecasts the following 24 hours; origins are anchored to local midnight, so each is a genuine day-ahead forecast. Accuracy is reported as WAPE, the total absolute error divided by total demand, which stays stable when many intervals carry near-zero demand, alongside MAE, RMSE and R^2. The interval is empirical: $\hat{y} \pm 1.645\,\sigma$, where σ is the residual standard deviation at the same horizon in the backtest, clipped at zero because demand cannot be negative.

4.4 Preliminary results

Table 2: Rolling-origin results, targets strictly out of sample. Boulder: 25 origins, 600 hours. Palo Alto: 14 origins, 336 hours. The interval is nominal 90 %.

Metric	Boulder (primary)	Palo Alto (validation)
WAPE, gradient boosting	0.292	0.549
WAPE, naive same hour yesterday	0.360	0.670
Relative reduction in WAPE	19 %	18 %
MAE (kWh per hour)	8.37	7.02
RMSE (kWh per hour)	11.39	9.70
R^2	0.763	0.466
Measured coverage of the 90 % interval	91.0 %	90.5 %

On the primary site the gradient-boosted model reaches a WAPE of 0.292 against 0.360 for the naive daily baseline, a reduction of 19 %, with R^2 of 0.763 and a measured interval coverage of 91.0 % against a nominal 90 %; the mean interval width is 33.2 kWh per hour. The same pipeline, with no parameter changed, reaches 0.549 against 0.670 on the validation site, still a reduction of 18 %, at 90.5 % coverage. The normalised error is higher on the validation site while the absolute error per hour is comparable (7.02 against 8.37 kWh), because the two windows differ in scale and in period; the absolute figures are therefore not comparable across sites, and the claim made here is the within-site relative gain, which repeats at 18 %. That repetition is the evidence that the gain comes from the method rather than from tuning. Figure 2 shows the last week of the evaluation window.

Two properties of the data constrain the platform that consumes these forecasts. Both sites peak in the middle of the working day, at 11:00 in Boulder and 12:00 in Palo Alto, and bottom out at 04:00 with a peak-to-trough ratio of about

nineteen; there is no residential evening peak, because this is public charging rather than home charging. A scheduler that assumes the evening peak of the project brief would therefore optimise the wrong hours, which is a finding the Phase-3 design has to carry.

5 References

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6 Measurable Outcomes and Work Plan

Table 3: Main tasks, measurable targets and dates. Type: S = software, D = data, A = analysis, T = theoretical.

Type	Task	Measurable target	Due
D	Acquire and register official session datasets	Two series reproducible from a checksummed download	Sep 2026
S	Session-to-hourly pipeline with contract IO	Gap-free hourly CSV plus metadata, $\geq$50 tests	Oct 2026
S	Feature set, direct multi-horizon models, backtest	WAPE reported against three naive baselines	Oct 2026
A	Interval construction and coverage check	Measured coverage within $\pm$3 pp of nominal	Nov 2026
A	Cross-site validation without retuning	Relative WAPE reduction $\geq$10 % on site 2	Nov 2026
S	Tariff-aware features and interface freeze	Contract v2 consumed by SP2 without change	Dec 2026
A	Interim report and presentation	Submitted 8 Jan; presented 11–22 Jan	Jan 2027
A	Phase-3 evaluation with SP2 and SP5	Peak, cost and CO_2 deltas quantified	Apr 2027
T	Dissertation and poster	Submitted 30 Apr; poster 19–21 Apr	Apr 2027

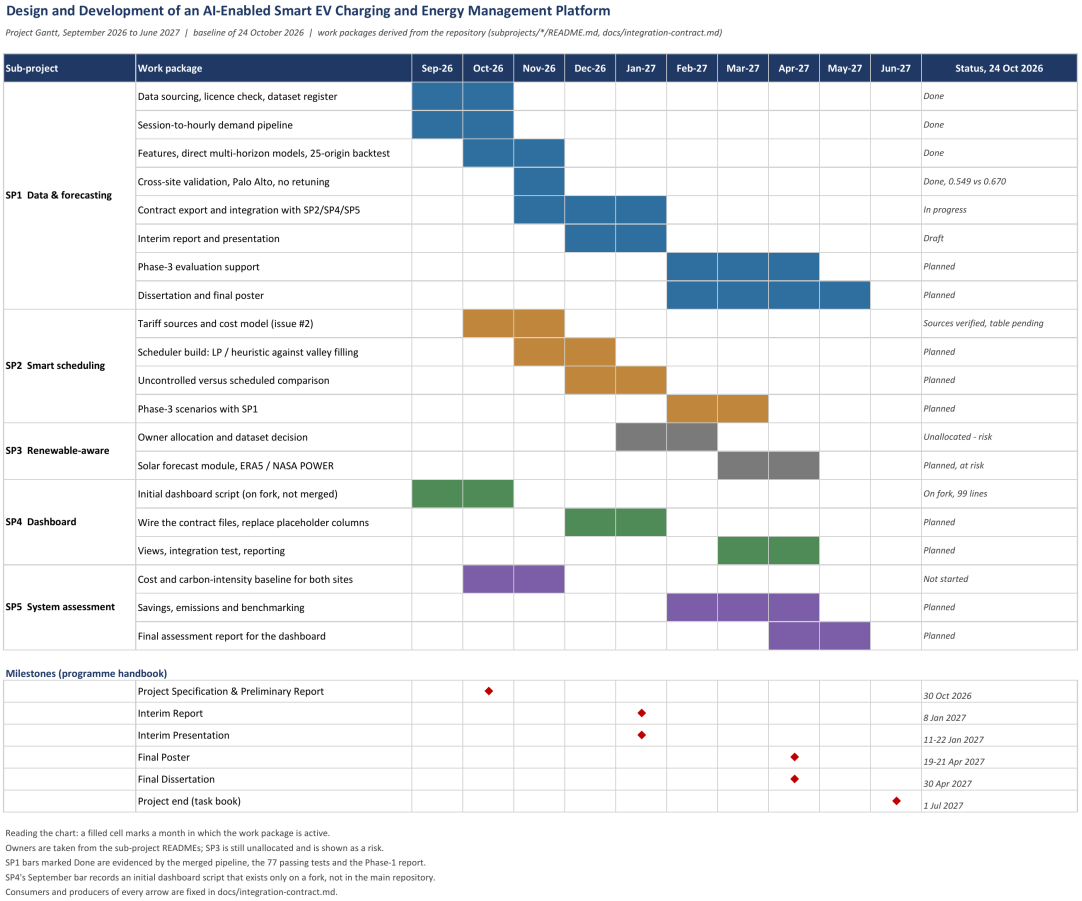


Figure 3: Project Gantt chart, September 2026 to June 2027. Work packages are derived from the sub-project definitions and the integration contract held in the repository; the workbook is supplied separately as `gantt.xlsx` so that the plan can be edited rather than redrawn.